Q3. [20 pts] The Singular Value Decomposition

We want you to compute (by hand) part of the singular value decomposition (SVD) $X = UDV^\top$ of the matrix

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix}.$$

(a) [6 pts] One way to compute the SVD by hand is to exploit the relationship between the singular values of X and the eigenvalues of a related symmetric matrix. What symmetric matrix has eigenvalues related to the singular values of X? Compute that matrix, write down its characteristic polynomial and simplify it, and determine its eigenvalues. Show your work!

(b) [4 pts] What is the relationship between the singular values of X and the eigenvalues of your symmetric matrix? Use that relationship to write down the two specific singular values of X.

(c) [5 pts] Generalize your answer above. For **any** matrix Z (not only X), what symmetric matrix has eigenvalues related to the singular values of Z? **Prove** that every singular value of Z can be identified by looking at the eigenvalues of the matrix your method constructs.

(d) [5 pts] Explicitly write down enough eigenvectors of your symmetric matrix to form a complete basis for the space. (Hint: the symmetry of your matrix should make it easy to guess eigenvectors; then you can confirm that they are eigenvectors.) Based on these, write down the value of U or V (whichever is appropriate).

(a) Consider the symmetric matrix

$$Y = X^\top X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

The characteristic polynomial of Y is $(\lambda - 2)^2 - 1 = \lambda^2 - 4\lambda + 3$. We can factor it into $(\lambda - 3)(\lambda - 1)$ or simply use the quadratic formula to see that its solutions are $\lambda = 3$ and $\lambda = 1$, the eigenvalues of Y.

Note: if you like doing things the hard way, it is also correct to use the matrix

$$XX^\top = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

(b) The singular values of X are square roots of eigenvalues of $X^\top X$ (or $XX^\top$). Therefore, they are $\sqrt{3}$ and 1.

(c) The singular values of Z are square roots of eigenvalues of $Z^\top Z$ (or $ZZ^\top$). Proof: let $Z = UDV^\top$ be the SVD of Z; then $Z^\top Z = VDU^\top UDV^\top = VD^2V^\top$, which is clearly an eigendecomposition (because V is orthonormal and D^2 is diagonal). If Z has a singular vector δ with right singular vector v, then v is an eigenvalue of $Z^\top Z$ with eigenvalue δ^2.

(d) The simplest basis of eigenvectors for $X^\top X$ is $[1, 1]^\top$ and $[1, -1]^\top$. From this we can choose

$$V = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$

but note that the $-1/\sqrt{2}$ could appear in any of the four slots. Moreover, we could have three negative entries instead of one.

Postscript: By the way, one correct SVD is

$$X = \begin{bmatrix} 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{6} & 1/\sqrt{2} \\ 2/\sqrt{6} & 0 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

Q4. [10 pts] Maximum Likelihood Estimation of a Mean

In class, we derived the maximum likelihood estimates for the mean and variance of a univariate (or isotropic multivariate) normal distribution, based on n points sampled independently from that distribution. Then we considered the more general (anisotropic) multivariate distribution $\mathcal{N}(\mu, \Sigma)$, and I told you the MLE estimates without deriving them. Let's derive the maximum likelihood estimate for the mean.

We are given n independent sample points $X_1, X_2, \ldots, X_n \sim \mathcal{N}(\mu, \Sigma)$ with d features each, where $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d \times d}$ is symmetric and positive definite.

(a) [4 pts] Write out an expression for the likelihood $\mathcal{L}(\mu, \Sigma; X_1, \ldots, X_n)$. (You should substitute in the normal PDF; a placeholder will only get part marks.) Simplify the expression as much as possible.

(b) [6 pts] Derive the maximum likelihood estimate $\hat{\mu}$ of the distribution mean μ. (You probably already know what its value should be. If you get it right, it will not depend on Σ.)

(a)

$$\mathcal{L}(\mu, \Sigma) = \frac{1}{(2\pi)^{nd/2}|\Sigma|^{n/2}} \prod_{i=1}^{n} \exp\left(-\frac{1}{2}(X_i - \mu)^\top \Sigma^{-1}(X_i - \mu)\right).$$

Alternatively,

$$\mathcal{L}(\mu, \Sigma) = \frac{1}{(2\pi)^{nd/2}|\Sigma|^{n/2}} \exp\left(-\frac{1}{2}\sum_{i=1}^{n}(X_i - \mu)^\top \Sigma^{-1}(X_i - \mu)\right).$$

(b) The log-likelihood is

$$\ell(\mu, \Sigma) = \ln \mathcal{L}(\mu, \Sigma) = -\ln\left((2\pi)^{nd/2}|\Sigma|^{n/2}\right) - \frac{1}{2}\sum_{i=1}^{n}(X_i - \mu)^\top \Sigma^{-1}(X_i - \mu).$$

Setting $\nabla_\mu \ell(\mu, \Sigma) = 0$ gives (because Σ^{-1} is symmetric)

$$\Sigma^{-1}\left(\sum_{i=1}^{n} X_i - n\mu\right) = 0.$$

Pre-multiplying both sides by Σ gives $\sum_{i=1}^{n} X_i - n\mu = 0$, hence $\hat{\mu} = \frac{1}{n}\sum_{i=1}^{n} X_i$, which is the sample mean.

Q5. [20 pts] Kernel Principal Components Analysis

In this problem, we will derive the kernelized version of principal components analysis (PCA). You are given a **centered** $n \times d$ design matrix X whose rows express the sample points $X_1^\top, X_2^\top, ..., X_n^\top$, and a feature map Φ that maps each sample point $X_i \in \mathbb{R}^d$ to a high-dimensional "lifted" sample point $\Phi(X_i) \in \mathbb{R}^D$, where $D \gg d$. Define the kernel function $k(X_i, X_j) = \Phi(X_i)^\top \Phi(X_j)$, and suppose it takes $O(d)$ time to compute (which is much, much less than D).

The sample covariance matrix of the lifted points in $\mathbb{R}^D$ is $C = \frac{1}{n}\Phi(X)^\top\Phi(X) = \frac{1}{n}\sum_{i=1}^n \Phi(X_i)\,\Phi(X_i)^\top \in \mathbb{R}^{D\times D}$. (Recall that $\Phi(X) \in \mathbb{R}^{n\times D}$. Recall also that our convention is to express each X_i and each $\Phi(X_i)$ as a column vector, but these points are stored as rows of X and $\Phi(X)$, respectively, which is why the transpose moved.)

(a) [6 pts] To perform PCA, we want to find the eigenvectors of C with nonzero eigenvalues. However, we don't ever want to construct C—it's way too big! We don't even want to construct one eigenvector of C! Show that if a vector $v \in \mathbb{R}^D$ is an eigenvector of C with a nonzero eigenvalue, then v can be expressed as a linear combination of lifted sample points: that is, $v = \sum_{j=1}^n a_j \Phi(X_j) = \Phi(X)^\top a$ for some vector $a \in \mathbb{R}^n$ of "dual" coefficients. Hint: think of the standard definition of an eigenvector of C, then substitute in the value of C.

(b) [5 pts] Recall that the kernel matrix $K = \Phi(X)\,\Phi(X)^\top$ is the symmetric $n \times n$ matrix such that $K_{ij} = k(X_i, X_j)$. In the circumstance described in part (a), prove that $K^2 a = \lambda n K a$. Hint: again start with the standard definition of an eigenvector of C, then substitute in the value of v from above.

(c) [5 pts] Conversely, suppose that a is an eigenvector of K with eigenvalue λ_a. Show that $v = \Phi(X)^\top a$ is an eigenvector of C. What is v's corresponding eigenvalue?

(d) [4 pts] The parts above imply that by finding the eigendecomposition of the kernel matrix K, we can indirectly express the eigenvectors (that have nonzero eigenvalues) of the kernelized covariance matrix C without ever explicitly computing C nor any vector of length D. Suppose that we have a test point $z \in \mathbb{R}^d$ and we want to efficiently compute the principal coordinate of $\Phi(z)$ projected onto a principal component $v = \sum_{j=1}^n a_j \Phi(X_j) = \Phi(X)^\top a$. Assume that we have already computed a. Describe how to compute $\Phi(z)$'s principal coordinate quickly, without ever computing a vector of length $\Theta(D)$ or doing a computation taking $\Theta(D)$ time. (Recall our assumption that kernel function computations are much, much faster than that.)

(a) As v is an eigenvector of C with a nonzero eigenvalue, there exists a nonzero scalar λ such that $Cv = \lambda v$. Substituting the formula for C into this expression gives us

$$\begin{aligned} \lambda v &= \frac{1}{n}\Phi(X)^\top\Phi(X)v \\ v &= \Phi(X)^\top \frac{1}{\lambda n}\Phi(X)v. \end{aligned}$$

So if we set $a = \frac{1}{\lambda n}\Phi(X)v$, the claim follows.

(b) Substituting $v = \Phi(X)^\top a$ and the formula for C into $Cv = \lambda v$ gives

$$\frac{1}{n}\Phi(X)^\top\Phi(X)\Phi(X)^\top a = \lambda\Phi(X)^\top a$$

Pre-multiplying both sides by $\Phi(X)$ gives

$$\begin{aligned} \frac{1}{n}\Phi(X)\Phi(X)^\top\Phi(X)\Phi(X)^\top a &= \lambda\Phi(X)\Phi(X)^\top a \\ K^2 a &= \lambda n K a. \end{aligned}$$

(c) As $\lambda_a a = Ka = \Phi(X)\Phi(X)^\top a$, we have $Cv = \frac{1}{n}\Phi(X)^\top\Phi(X)\Phi(X)^\top a = \frac{1}{n}\Phi(X)^\top Ka = \frac{\lambda_a}{n}\Phi(X)^\top a = \frac{\lambda_a}{n}v$. So v is an eigenvector of C with eigenvalue $\frac{\lambda_a}{n}$.

(d) The principal coordinate of $\Phi(z)$ on principal component v is $\Phi(z)^\top v$, so

$$\Phi(z)^\top \sum_{j=1}^{n} a_j \Phi(X_j) = \sum_{j=1}^{n} a_j k(z, X_j).$$